

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-ada44152b85badf28.jsonl`  
- SHA-256: `0dddc6cce0739b2e7ac09079593b649106f4c060f2b2238028dfcdcf5a3cd6d0`  
- Source modified: `2026-06-16T15:14:30+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_736be611-fb5`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T15:12:42.309Z)

You are a small-data evaluation-rigor expert. The owner argues strict tIoU 0.5 over-penalizes because the SERVICE WINDOW makes rally starts intrinsically ~1-1.5s fuzzy (even Gemini's median |?start| is 1.46s), and proposes a +-2s tolerance. Critique the current metric set (tIoU-F1, F1@k, merge/split, mIoU) for THIS task. Design the task-faithful metric: a +-tau tolerance match, boundary-error-in-seconds split start/end, while KEEPING an over-segmentation guard so a relaxed tolerance cannot hide merges. What tau, and how does it relate to inter-annotator agreement? At n=6-7 what makes a delta trustworthy (CI + sign-test)? Cite numbers.

REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI WASB windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
GX010128: 39 | 15 | 78 | 75
GX010129: 27 | 2 | 50 | 47
GX020125: 11 | 6 | 18 | 15
mahadevpura-0: 40 | 19 | 62 | 43
mahadevpura-1: 9 | 1 | 2 | 2
mahadevpura-2: 39 | 5 | 40 | 40
AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity gap),
mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
F1@0.5 / F1@0.25 / mean-IoU: Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
=> local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service window
nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139, 6/0)
-> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on n=7
(continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift +

sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity > velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 = max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap = midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

For raw per-video detail you MAY Read C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json . Treat the DATA block as authoritative for the numbers. Constraints: local pipeline must stay MIT/Apache/BSD-only, single local GPU, ~6-7 golden videos now (~16 by the weekend), Gemini usable at inference only (never as a training-data source).

2. assistant (2026-06-16T15:12:46.032Z)

I'll analyze the data and design a task-faithful metric. Let me start by reading the raw per-video detail to ground my critique in the actual numbers.

3. user (2026-06-16T15:12:47.665Z)

```
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5          "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-
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56     "name": "GX010129",
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61     "mean_gem": 7.2444444444444445,
62     "variants": {
63         "baseline": {
64             "n": 2,
65             "play": 534.4177510844178,
66             "mean_dur": 267.2088755422089,
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167     "mean_gem": 6.55,
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224             "play": 174.2742742742743,
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235         },

```

```

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251         "n": 2,
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310         "local_only": 4,
311         "gemini_only": 0,
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315         "gem_rallies_in_merges": 11
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317 },
318 "status": "ok"
319 }
320 ]

```

4. assistant (2026-06-16T15:13:01.959Z)

I have everything I need. The DATA block plus the per-video JSON are sufficient ? this is an evaluation-design question, not a web-research one. Let me reason it through and return the structured analysis.

A few key derivations I'll anchor on, computed from the data:

****IoU?time conversion**** (for a rally of duration D, a pure translation by ? with equal-length prediction gives $\text{IoU} = (D??)/(D+?)$):

- Solving $\text{IoU} = 0.5$? ? = $D/3$. For D = 5.7s (owner GT mean): ? = 1.9s. For D = 6.82s (Gemini mean on mahadevpura-2): ? = 2.27s.

- Solving $\text{IoU} = 0.25$? ? = $0.6 \cdot D = 3.42\text{s}$ at $D=5.7\text{s}$.

- So $\text{tIoU } 0.5$? a $\pm 1.9\text{s}$ tolerance and $\text{tIoU } 0.25$? a $\pm 3.4\text{s}$ tolerance ? *for translation only*. For a symmetric shrink/grow it's even tighter.

****The diagnostic split**** (from GROUND-TRUTH): local |?start| median 1.74s ? Gemini's 1.46s, but local |?end| median 3.12s vs Gemini's 1.19s. A single tIoU number blends these; it cannot show that start is at-par and end is the entire gap. This is the central indictment of the current metric.

****Over-segmentation is the live failure mode, not merging**** (w1+w2 count_ratio: GX010128 1.92, GX010129 1.74, GX020125 1.36, mahadevpura-0 1.075, mahadevpura-2 1.03; aggregate 222 local vs 165 Gemini = 1.35). A pure tolerance match with greedy 1:1 assignment would let these extra predictions hide for free ? that's exactly the trap to guard against.

Let me also verify the sign-test arithmetic for n=6 and n=7 before I commit it to recommendations.

5. user (2026-06-16T15:13:04.628Z)

n= 5

```

k=0/5  two-sided p=1.0000
k=1/5  two-sided p=1.0000
k=2/5  two-sided p=1.0000
k=3/5  two-sided p=1.0000
k=4/5  two-sided p=0.3750
k=5/5  two-sided p=0.0625

```

n= 6

```

k=0/6  two-sided p=1.0000
k=1/6  two-sided p=1.0000
k=2/6  two-sided p=1.0000
k=3/6  two-sided p=1.0000

```

```

k=4/6  two-sided p=0.6875
k=5/6  two-sided p=0.2188
k=6/6  two-sided p=0.0312
n= 7
k=0/7  two-sided p=1.0000
k=1/7  two-sided p=1.0000
k=2/7  two-sided p=1.0000
k=3/7  two-sided p=1.0000
k=4/7  two-sided p=1.0000
k=5/7  two-sided p=0.4531
k=6/7  two-sided p=0.1250
k=7/7  two-sided p=0.0156

```

6. assistant (2026-06-16T15:13:10.102Z)

The sign-test arithmetic confirms the small-n reality: at n=6 you need a clean 6/0 sweep to even reach p=0.031 (one-sided ~0.016); at n=7 you need 7/0 (two-sided p=0.0156). This is decisive for what "trustworthy" can mean. The prior golden deltas being "6/0" is exactly why baseline?W1 (+0.139) was trustworthy while W1?W1+W2 (+0.019, also 6/0 but tiny effect) is direction-trustworthy but magnitude-negligible.

Let me also verify the IoU?time relationship and the over-segmentation precision penalty arithmetic that anchors my guard design.

7. user (2026-06-16T15:13:16.327Z)

```

D=5.70 tIoU=0.50 -> max translation delta=1.90 s
D=5.70 tIoU=0.25 -> max translation delta=3.42 s
D=6.82 tIoU=0.50 -> max translation delta=2.27 s
D=6.82 tIoU=0.25 -> max translation delta=4.09 s
D=5.69 tIoU=0.50 -> max translation delta=1.90 s
D=5.69 tIoU=0.25 -> max translation delta=3.41 s
---over-seg precision under tolerance matching---
npred=40 nref=40 matched=12 -> P=0.300 R=0.300
aggregate count ratio local/gem = 1.3454545454545455

```

8. assistant (2026-06-16T15:13:18.939Z)

All arithmetic confirmed. I have everything needed to deliver the critique and the metric design.

9. user (2026-06-16T15:14:16.556Z)

Structured output provided successfully

10. assistant (2026-06-16T15:14:30.167Z)

I've delivered the critique and metric design via StructuredOutput. Summary of the analysis:

****The owner is right (arithmetically):**** tIoU 0.5 tolerates only ? = D/3 of pure boundary shift ? 1.90s at the GT mean rally (D=5.7s). The service-window start fuzz (Gemini median |?start|=1.46s) alone consumes ~0.77?1.0 of that budget before any end error, so tIoU 0.5 functionally is a sub-2s tolerance that the irreducible ambiguity nearly exhausts.

****Core indictment of the current set:**** a single tIoU/mIoU blends start and end error, hiding the one fact that matters ? local START is already at Gemini parity (1.74s ? 1.46s) and the *entire* gap is rally-END (3.12s vs 1.19s). The metric can't direct the next fix.

****The task-faithful metric:****

- ****Asymmetric tolerance match:**** |?start| ? ?_start=2.0s AND |?end| ? ?_end=1.0?1.5s. Start gets 2× the inter-annotator noise floor (service window is fuzzy); end stays tight because rally-end is a sharp event and is exactly where local fails.
- ****Over-segmentation guard** (so a relaxed ? can't hide merges):** strict 1:1 global (Hungarian) assignment, report Precision-at-? alongside Recall-at-?. The 222/165 = 1.35 surplus then costs

precision directly and can't match for free.

- ****Fragmentation/merge pair**** as a hard no-regression gate (catches both W1 over-seg and the baseline mega-window pathology).
- ****Boundary error reported as two separate signed+absolute distributions****, never folded into IoU.

****Trustworthiness at n=6-7 (sign-test is binding, not CI):**** n=6 needs a clean 6/0 sweep for $p=0.031$ (5/1 ? $p=0.219$); n=7 needs 7/0 for $p=0.016$ (6/1 ? $p=0.125$). A bootstrap mean-F1 CI at this n spans $\sim \pm 0.15 \text{ } 0.20$ and will overlap zero ? report it as an honesty band, not a gate. This is why baseline?W1 (+0.139, 6/0) is real while W1?W1+W2 (+0.019, 6/0) is directionally-clean-but-negligible.

Relevant file: `C:\Users\avidu\Projects\badminton-highlight-indexer\output\local\_vs\_gemini\_6.json` (per-video matched/merge/split/count_ratio detail) and the tracked doc to record ? + the IoU?time derivation: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN\_REGRESSION\_FIXTURES.md`.